METRIC
MIL-F-24702(SH)
AMENDMENT 2
28 November 1994
SUPERSEDING
AMENDMENT 1
2 MAY 1991

MILITARY SPECIFICATION

FILTER ELEMENTS, HYDRAULIC DISPOSABLE,
GENERAL SPECIFICATION FOR (METRIC)

This amendment forms a part of MIL-F-24702(SH) dated 28 April 1988, and is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

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2.1.1 Specifications and standards: Add the following standard:

"MIL-STD-2193 - Hydraulic System Components, Ship. Metric"

* 2.1.1 Specifications and standards: Delete the following standard:

"MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes"

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* 2.2 Other Publications: Delete the following:

"B93.31 - Mult-Pass Method for Evaluating the Filtration Performance of a Fine Hydraluic Fluid Power Filter Element."

* 2.2 Other publications: Add the following, after the "ANSI" address for copies:

"NATIONAL FLUID POWER ASSOCIATION (NFPA)

T.3.10.8.8 - Multi-Pass Method for Evaluating the Filtration Performance of a Fine Filters.

(Applications for copies should be addressed to the National Fluid Power Association, 3333 N. Mayfair Road, Milwaukee, WI 53222-3219)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (IS0)

IS0 4406 - Method for Coding Level of Contamination by Solid Particles, ″

* 2.2 Other publications: Delete the following:

" AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC.

NAS 1638 - Cleanliness Requirements of Parts Used i Hydraulic System."

(Applications for copies should be addressed to the Aerospace Industries Association of America, Inc. 1725 DeSales Street, NW, Washington, DC 20036.)"

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3.3.3: Delete and substitute:

"3.3.3 Seals and element connectors. Seals and seal glands shall be in accordance with MIL-STD-2193. Non-metallic materials, except for those specifically listed in MIL-STD-2193, may be used only when verified to be compatible by conducting the immersion test in MIL-STD-2193 for plastic parts."

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TABLE I: Add the following:

Non-metallic seals and element connectors		
Examination Compatibility	3.3.3, 3.8 3.3.3, 3.8	4.6 4.7.7.1

TABLE I: Delete "3.9" and substitute "3.8" in four locations.

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4.4: Delete and substitute:

"4.4 Quality conformance inspection."

* 4.4.1: Delete and substitute:

"4.4.1 Quality conformance sampling. Sampling for quality conformance inspection shall be as specified (see 4.4.1.1, 4.4.1.2 and 4.4.1.3 and 4.4.1.4)."

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4.4.1.1 Sampling for examination and tests. For the purpose of sampling, filter elements shall be grouped into lots not exceeding 501 pieces. A lot is defined as filters produced at one facility, fabricated by the same process, which are of the same material, size, type and part number.

4.4.1.2 Examination and bubble point. As a minimum the manufacturer shall select a sample quantity from each lot of filter elements in accordance with table II-1 and inspect them in accordance with table II-2. If one or more defects are found in any sample, the entire lot shall be rejected. The manufacturer has the option of screening 100 percent of the rejected lot for the defective characteristic(s) or providing a new lot which shall be inspected in accordance with the sampling plan contained herein.

4.4.1.3 Clean element pressure drop. As a minimum the manufacturer shall select a sample quantity from each lot of filter elements in accordance with table II-1 and inspect them in accordance with table II-2. Failure of the test shall be cause for rejection of the element lot. The manufacturer has the option of screening 100 percent of the rejected lot for the defective characteristic(s) or providing a new lot which shall be inspected in accordance with the sampling plan contained herein.

* 4.4.1.4: Add as new paragraph:

"4.4.1.4 Filter element performance. For this test, sampling of a filter element from each lot is not normally required. One element shall be selected from the first lot produced for a procurement, This element shall be subjected to the filtration ratio, dirt holding, capacity and pressure drop test in accordance with 4.7.2. If this element passes all the tests, all lots of this particular configuration element produced within 10 consecutive days of the production of the test element may be accepted. If the element tested fails to meet the requirements , three additional filter elements from the same lot shall be tested and if all three additional filter elements pass the test, the lot may be accepted. Whenever an elements fails to pass the test, all lots for a particular procurement shall be sampled until four lots are accepted without any elements failing to meet the test.

* TABLE II-1: Delete and substitute:

TABLE II-1. Sampling for quality conformance inspection.

Lot size	Sample size		
	Examination and bubble point	Clean element pressure drop	Filtration Performance
1 - 29	All	1	*
30 - 501	29	1	1*

* See 4.4.1.4

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* TABLE II: Delete and substitute:

TABLE II-2. Quality conformance inspection.

Examinations	Requirement	Test
(1) Examination	3.3, 3.4.1, 3.3, 3.7 and 3.8	4.6
(2) Bubble point	3.5.1	4.7.1
(3) Clean element pressure drop	3.5.2	4.7.2
(4) Filter performance	3.5.2	4.7.2

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* 4.5.1.1, Line 3, delete "4 in accordance with NAS 1638" and substitute "IS0 Code */12/10."

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* 4.7.2, Line 2, delete "ANSI B93.31" and substitute "ANSI/NFPA T3.10.8.8".

4.7.3, line 4: Delete "the maximum specified" and substitute "60 lb/in^2 minimum".

4.7.3: Add the following sentence:

"If the element bubble point drops below 80 percent, then, first clean the element with solvent; and perform the filtration ratio test at terminal pressure only to demonstrate element ability of meeting the beta ratio requirement."

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Add as new paragraph 4.7.7.1:

"4.7.7.1 Seal and element connector compatibility. Two samples of each non-metallic seal and element connector shall be subjected to the immersion test specified in MIL-STD-2193."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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